

# Spatial Transcriptomics Cheatsheet

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The tables below consist of valuable functions or commands that will help you through this module.

Each table represents a different library/tool and its corresponding commands.

Please note that these tables are not intended to tell you all the information you need to know about each command.

The hyperlinks found in each piece of code will take you to the documentation for further information on the usage of each command. Please be aware that the documentation will generally provide information about the given function's most current version (or a recent version, depending on how often the documentation site is updated). This will usually (but not always!) match what you have installed on your machine. If you have a different version of R or other R packages, the documentation may differ from what you have installed.

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## SpatialExperiment

Read the [SpatialExperiment package documentation \(and e-book\)](#) and a [vignette on its usage](#).

`SpatialExperiment` objects are based on `SingleCellExperiment` objects. There are many functions that are shared between the two, with equivalent usage. See [this SingleCellExperiment vignette](#) for example usage of shared functions. You can also refer to the Data Lab cheatsheets for our "Introduction to scRNA-seq" ( `scRNA-seq-cheatsheet.pdf` ) and "Advanced scRNA-seq" ( `scRNA-seq-advanced-cheatsheet.pdf` ) training modules.

| Library/Package                | Piece of Code                    | What it's called               | What it does                                                                       |
|--------------------------------|----------------------------------|--------------------------------|------------------------------------------------------------------------------------|
| <code>SpatialExperiment</code> | <code>SpatialExperiment()</code> | Spatial Experiment             | Creates a <code>SpatialExperiment</code> object                                    |
| <code>SpatialExperiment</code> | <code>spatialCoords()</code>     | Spatial Experiment coordinates | Extracts spatial coordinates of spots in the <code>SpatialExperiment</code> object |
| <code>SpatialExperiment</code> | <code>imgData()</code>           | Spatial Experiment images      | View the image data associated with a <code>SpatialExperiment</code> object        |

## VisiumIO

Read the R package [VisiumIO documentation](#) and a [vignette on its usage](#).

| Library/Package       | Piece of Code             | What it's called | What it does                                                                          |
|-----------------------|---------------------------|------------------|---------------------------------------------------------------------------------------|
| <code>VisiumIO</code> | <code>TENxVisium()</code> | 10x Visium       | Constructor function for importing 10x Visium data                                    |
| <code>VisiumIO</code> | <code>import()</code>     | Import           | Loads 10x Visium data defined with the <code>TENxVisium()</code> constructor function |

## ggspavis

Read the [ggspavis package documentation](#) and a [vignette on its usage](#).

| Library/Package       | Piece of Code             | What it's called | What it does                                                                                                                                                                           |
|-----------------------|---------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ggspavis</code> | <code>plotCoords()</code> | Plot coordinates | Creates a spot plot showing spatial locations in x/y with options for spot annotation                                                                                                  |
| <code>ggspavis</code> | <code>plotVisium()</code> | Plot Visium      | Creates a spot plot for Visium data specifically showing spatial locations in x/y with options for spot annotation as well as an option to show the spatial image, e.g., the H&E image |

## SpotSweeper

Read the [SpotSweeper package documentation](#) and a [vignette on its usage](#).

| Library/Package | Piece of Code                | What it's called | What it does                                                                                           |
|-----------------|------------------------------|------------------|--------------------------------------------------------------------------------------------------------|
| SpotSweeper     | <code>localOutliers()</code> | Local outliers   | Detect local outlier spots in a <code>SpatialExperiment</code> based on a given quality control metric |
| SpotSweeper     | <code>plotQCmetrics()</code> | Plot QC metrics  | Create a spot plot highlighting outliers detected with <code>SpotSweeper::localOutliers()</code>       |

## Banksy

Read the [Banksy package documentation](#), which contains several vignettes on its usage, including a [vignette on parameter selection](#). Note that many of their vignettes use of image-based spatial transcriptomics data that is more fine-grained compared to 10x Visium.

| Library/Package | Piece of Code                | What it's called            | What it does                                                                             |
|-----------------|------------------------------|-----------------------------|------------------------------------------------------------------------------------------|
| Banksy          | <code>computeBanksy()</code> | Compute <code>Banksy</code> | Compute the component neighborhood matrices for the <code>Banksy</code> matrix           |
| Banksy          | <code>runBanksyPCA()</code>  | Run <code>Banksy</code> PCA | Run PCA on a <code>Banksy</code> matrix while weighting the spatial neighborhood context |

## spacexr

Read the [spacexr package documentation](#), and a [vignette on using their RCTD algorithm](#).

| Library/Package | Piece of Code                     | What it's called                  | What it does                                                                  |
|-----------------|-----------------------------------|-----------------------------------|-------------------------------------------------------------------------------|
| spacexr         | <code>createRctd()</code>         | Create an RCTD object             | Preprocess data before performing deconvolution with RCTD                     |
| spacexr         | <code>runRCTD()</code>            | Run RCTD                          | Run the RCTD algorithm to decompose cell type mixtures                        |
| spacexr         | <code>plotCellTypeWeight()</code> | Plot a single cell type's weights | Plot pixel proportions for a specific cell type on a spatial coordinates plot |

## scuttle, scran, and scater

Read the [scuttle package documentation](#), and a [vignette on its usage](#).

Read the [scran package documentation](#), and a [vignette on its usage](#).

Read the [scater package documentation](#), and a [vignette on its usage](#).

| Library/Package | Piece of Code                        | What it's called              | What it does                                                                                                                                                                                                                                                                                         |
|-----------------|--------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| scuttle         | <code>addPerCellQC()</code>          | Add per cell quality control  | For a <code>SingleCellExperiment</code> object, calculate and add quality control per cell and store in <code>colData</code>                                                                                                                                                                         |
| scuttle         | <code>computeLibraryFactors()</code> | Compute Library Factors       | Returns a numeric vector of computed size factors for each spot (or cell) stored in a <code>SpatialExperiment</code> (or <code>SingleCellExperiment</code> ) object. The size factor is computed as the library size of each spot/cell after scaling them to have a mean of 1 across all spots/cells |
| scuttle         | <code>logNormCounts()</code>         | Normalize log counts          | Returns the <code>SpatialExperiment</code> (or <code>SingleCellExperiment</code> ) object with normalized expression values for each spot (cell), using the size factors stored in the object                                                                                                        |
| scran           | <code>getTopHVGs()</code>            | Get top highly variable genes | Identify variable genes in a <code>SingleCellExperiment</code> object, based on variance                                                                                                                                                                                                             |
| scran           | <code>modelGeneVar()</code>          | model per gene variance       | Model the per gene variance of a <code>SingleCellExperiment</code> object                                                                                                                                                                                                                            |
| scran           | <code>clusterCells()</code>          | Cluster cells                 | Perform clustering on an SCE object using the <code>bluster</code> package                                                                                                                                                                                                                           |
| scater          | <code>runPCA()</code>                | Run PCA                       | Calculates principal components analysis on a <code>SingleCellExperiment</code> object, returning an SCE object with a PCA reduced dimension                                                                                                                                                         |
| scater          | <code>runUMAP()</code>               | Run UMAP                      | Calculates uniform manifold approximate projection on a <code>SingleCellExperiment</code> object, returning an SCE object with a UMAP reduced dimension                                                                                                                                              |
| scater          | <code>plotUMAP()</code>              | Plot UMAP                     | Plot the "UMAP"-named reduced dimension slot from a <code>SingleCellExperiment</code> object                                                                                                                                                                                                         |

## bluster

Read the [bluster package documentation](#) and vignettes on its usage:

- [Flexible clustering for Bioconductor](#)
- [Assorted clustering diagnostics](#)

| Library/Package | Piece of Code                     | What it's called                          | What it does                                                                                                                                                  |
|-----------------|-----------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bluster         | <code>NNGraphParam()</code>       | Graph-based clustering parameters         | Set up parameters for nearest-neighbor (NN) graph-based clustering algorithms within <code>scrn::clusterCells()</code> or <code>bluster::clusterRows()</code> |
| bluster         | <code>approxSilhouette()</code>   | Approximate silhouette width              | Calculate an approximate silhouette width for each cell given a set of clusters                                                                               |
| bluster         | <code>neighborPurity()</code>     | Compute neighborhood purity               | Calculate neighborhood purity for each cell given a set of clusters                                                                                           |
| bluster         | <code>bootstrapStability()</code> | Assess cluster stability by bootstrapping | Generate cluster bootstrap replicates to estimate cluster robustness to sampling noise                                                                        |

## patchwork

Read the [patchwork package documentation](#), and a [vignette on its usage](#).

| Library/Package | Piece of Code             | What it's called | What it does                                                                                                                                                                     |
|-----------------|---------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| patchwork       | <code>wrap_plots()</code> | Wrap plots       | Wrap multiple <code>ggplot2</code> objects into a single multi-panel plot                                                                                                        |
| patchwork       | <code>+</code>            | Plot arithmetic  | Place <code>ggplot2</code> objects side-by-side by adding them together with a <code>+</code> . The <code>patchwork</code> package must be loaded to use this symbol with plots  |
| patchwork       | <code>/</code>            | Plot arithmetic  | Stack <code>ggplot2</code> objects on top of one another "dividing" them with a <code>/</code> . The <code>patchwork</code> package must be loaded to use this symbol with plots |